Polar PR-400 compressor rack: replacement of start capacitor assembly

Product family	Polar compressor racks
Models	PR-400, PR-400T (tropical variant)
Change class	Class B - field retrofit at next scheduled service visit
Originator	Ingrid Solheim, Product Engineering Manager, Polar
Change requested by	Quality (CAPA-2024-219) following field failure trend
Effectivity (production)	Immediate, all plants, from production week 2025-W09
Supersedes	ECN-2025-014 Rev A (27 Jan 2025)

1. Reason for change

Between October and December 2024 the field service desk logged 17 failures of the compressor start capacitor on Polar 400 racks, all within 14 months of commissioning. Typical symptom: compressor fails to start after a defrost cycle, rack alarm "C1 start fault", store loses cooling on the connected cases until a technician attends. Teardown of 9 returned parts by Quality found electrolyte leakage consistent with an under-rated component batch from the supplier.

Root cause (CAPA-2024-219): capacitor KC-3545-A from Kestrel Components s.r.o., lot codes K24-07 to K24-19, does not meet the 85 °C rating stated on the data sheet when installed in the rack's upper electrical enclosure. The tropical variant runs hotter and is the most exposed.

2. Description of change

	Before	After
Part number	KC-3545-A	VE-450-35M
Description	Start capacitor 35 uF 450 V, 85 °C	Start capacitor 35 uF 450 V, 105 °C
Supplier	Kestrel Components s.r.o. (CZ)	Vantor Electric GmbH (DE)
Bracket	BRK-PR-112	BRK-PR-112 (unchanged)
Unit cost	EUR 6.40	EUR 11.90

The mounting bracket and wiring harness are unchanged. The replacement part is a direct drop-in. Kestrel KC-3545-A remains approved for the PR-250 rack, which uses a lower-temperature enclosure and is not affected by this notice.

3. Affected items

Units built at the Trondheim plant between production weeks 2024-W10 and 2024-W35, serial numbers PR400-2403-0001 through PR400-2408-0412 inclusive. This includes the PR-400T variant, which carries the same serial sequence. Rev A of this notice gave the upper bound as PR400-2408-0380; Quality extended it after confirming lot K24-19 was used until the end of week 35.

Units built at the Brno plant in the same period used capacitor lots from a different supplier shipment (K23 lots) and are **not** affected, even where their serial numbers fall inside the range above. Brno-built units can be identified by plant code BRN on the rating plate and in ERP.

Exception: units PR400-2407-0350 through PR400-2408-0412 shipped to customers in the United Kingdom were fitted with VE-450-35M before dispatch under production deviation D-112. They do not require the retrofit. Service should confirm the fitted part on site before closing the work order.

4. Disposition

Stage	Action
Stock (all plants)	Quarantine KC-3545-A lots K24-07 to K24-19. Return to supplier under RMA-KC-0441.
Work in progress	Rework before final test. Record ECN number on the build traveller.
Finished goods	Rework before dispatch.
Field units	Retrofit at the next scheduled service visit, and in any case by 30 Sep 2025. Earlier if the unit raises a C1 start fault.

5. Service instructions

Service work orders for the retrofit must use activity code RET-014 and reference this ECN. Estimated labour is 0.75 h per rack plus travel. Technicians should replace the capacitor on both compressor stages where two are fitted (PR-400T always has two).

Where the retrofit is combined with a scheduled preventive maintenance visit under a service contract, only the preventive maintenance time is booked to the contract. Retrofit labour and parts are booked to the warranty cost centre WC-ENG-2025. **Do not book RET-014 time to the customer's service contract.**

6. Cost and warranty

Estimated field population: 380 to 412 units, less the UK units built under D-112. Estimated cost of the field action: parts EUR 12 per unit (two capacitors on the T variant), labour and travel EUR 140 to 260 per visit depending on country. A supplier recovery claim has been raised against Kestrel Components (SRC-2025-006). Warranty terms for affected units are extended by 12 months for the compressor electrical assembly only.

7. References

CAPA-2024-219; production deviation D-112; RMA-KC-0441; supplier recovery claim SRC-2025-006; Service Bulletin SB-PR-2025-02 (customer-facing summary, issued separately).

Revision history

Rev	Date	Change
A	27 Jan 2025	First issue. Affected range PR400-2403-0001 to PR400-2408-0380.
B	18 Feb 2025	Range extended to PR400-2408-0412. UK exception under D-112 added. Brno exclusion clarified.

Approvals

Role	Name	Signed	Date
Product Engineering	Ingrid Solheim	Signed	17 Feb 2025
Quality	Tomas Hrubec	Signed	17 Feb 2025
Service Operations	Marte Lien	Signed	18 Feb 2025
Supply Chain	Henrik Aasen	Signed	18 Feb 2025

FICTIONAL DOCUMENT - Datavid candidate assignment starter pack